

PAPER • OPEN ACCESS

Layer-by-layer deposition of nanostructured CsPbBr₃ perovskite thin films

To cite this article: A A Reshetnikova *et al* 2017 *J. Phys.: Conf. Ser.* **917** 052023

View the [article online](#) for updates and enhancements.

You may also like

- [High-performance thin-film photodetector based on lead-free perovskite CsBi_{1-x}/C8-BTBT heterojunction](#)
Ziyi Yang, Tengxiao Xionsong, Yuguang Luo et al.
- [Change in the electronic structure of the bismuth chalcogenide superconductor CsBi_{1-x}Pb_xTe₂ by dissociation of the bismuth dimers](#)
Hiroyuki Okazaki, Kensei Terashima, David Billington et al.
- [To study the structural, electronic and optical properties of predicted stable halide perovskites ABX₃](#)
Kashif Murad, Muhammad Noman, Shamim Khan et al.



The banner features a large white circle on the left containing the number '250' in red and green, with a blue ribbon below it that says 'ECS MEETING CELEBRATION'. To the right of the circle, the ECS logo is displayed above the text 'The Electrochemical Society' and 'Advancing solid state & electrochemical science & technology'. The background is a collage of confetti and people celebrating. On the right side, there is a green box with the text 'Step into the Spotlight' in white script. Below this, a red button with white text says 'SUBMIT YOUR ABSTRACT'. At the bottom right, the text 'Submission deadline: March 27, 2026' is written in blue.

250th ECS Meeting
October 25–29, 2026
Calgary, Canada
BMO Center

ECS The Electrochemical Society
Advancing solid state & electrochemical science & technology

*Step into the
Spotlight*

**SUBMIT YOUR
ABSTRACT**

**Submission deadline:
March 27, 2026**

Layer-by-layer deposition of nanostructured CsPbBr₃ perovskite thin films

A A Reshetnikova, L B Matyushkin, A A Andronov, V S Sokolov, O A Aleksandrova, V A Moshnikov

Department of micro- and nanoelectronics, St. Petersburg Electrotechnical University "LETI", St. Petersburg 197376, Russia

Abstract. Layer-by-layer deposition of nanostructured perovskites cesium lead halide thin films is described. The method of deposition is based on alternate immersion of the substrate in the precursor solutions or colloidal solution of nanocrystals and methyl acetate/lead nitrate solution using the device for deposition of films by SILAR and dip-coating techniques. An example of obtaining a photosensitive structure based on nanostructures of ZnO nanowires and layers of CsPbBr₃ nanocrystals is also shown.

1. Introduction

The development of solar cells with an active layer of a thin perovskite film became one of the most promising technology for the solar energy application in the past several years [1]. Perovskites (general chemical formula ABX₃) allow increasing the power conversion efficiencies larger than 20% [1, 2]. The technology of perovskite thin film synthesis is relatively simple: a standard technique is the spin-coating of a solution containing A, B, X elements, then desiccation in a drying cabinet. For example, CH₃NH₃PbI₃ films usually sensitized from CH₃NH₃I and PbI₂ precursors soluble in the polar fluids, such as dimethylformamide or dimethyl sulfoxide.

However, the thermal and chemical instability of the hybrid (organic-inorganic) perovskites leads to a search for similar structure and efficiency, but more suitable materials. In the perovskite materials group, it can be considered a family of lead cesium halides CsPbX₃ (X = Cl, Br, I). The use of cesium Cs instead or in addition to methylammonium group CH₃NH₃ allows obtaining dramatically more stable structures [3].

Materials of this group in the form of nanocrystals (NCs) and nanorods also have an effective photo- and electroluminescence [4]. The halogen type defines the band gap, and the luminescence spectra position: CsPbCl₃, CsPbBr₃ and CsPbI₃ NCs emit in the blue, green and red regions respectively. So, in the case of CsPbX₃ materials, it is possible to achieve fine tuning of absorption edge and luminescence maximum in the whole visible spectral range as a result of the selection of CsPbX₃ solid solution composition or nanocrystals size [5]. It is noteworthy that CsPbX₃ perovskite NCs have the cube shape, which facilitates better carrier transport conditions compared with commercial CdSe/ZnS QDs having a spherical shape.

Thus, using CsPbX₃ system one can create light emitting devices and solar cells contemporaneously. In the case of creating solar cells, it is possible to use FTO or ITO as transparent conductive oxide, TiO₂ or ZnO as electron transport layer; CsPbX₃ as perovskite layer (CsPbI₃ in the case of solar cells), CuO or NiO as hole transport layer, and Au or Ag as a back electrode.

The above-described spin-coating method's hard to scale and leads to material losses during the deposition process. In this work, we proposed solution-based perovskite deposition technique by



immersion of substrates in the precursor solutions or colloidal solutions of NCs using the device used for SILAR technique deposition of films [6, 7].

2. Description of the method

The device for the deposition of thin films is represented in Figure 1. It has horizontal and vertical actuators transferring the substrate holder respect from one thermostable beaker with a solution to another by predetermined synthesis parameters. The solutions are heated to temperatures controlled by detectors in chemically inert shells of stainless steel. The whole device is operated by Arduino controller connected to a computer with a program designed in NI LabVIEW environment using LabVIEW Interface for Arduino. An operator sets in the program the technological parameters of thin film deposition: the number of cycles, volume, and temperature of precursors, the dimensions of the substrate, the delay times between operations, etc. [7].

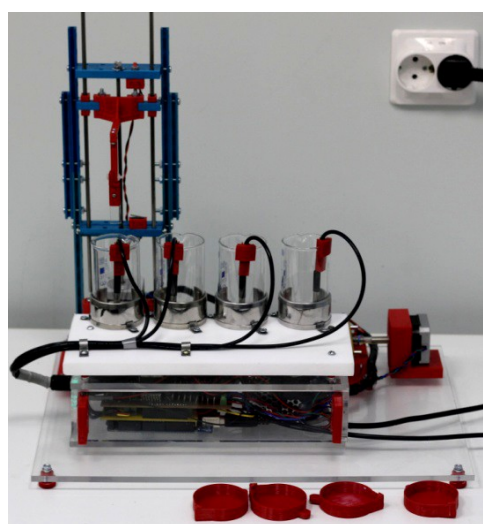


Figure 1. The device for the deposition of thin films

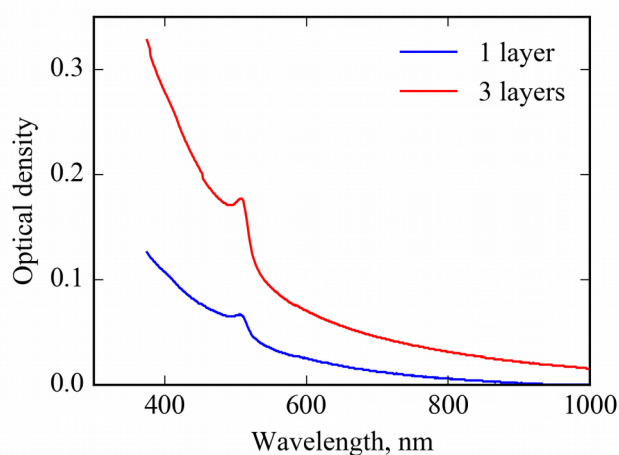


Figure 2. Optical density spectra of thin films after 1 and three cycles of layering of CsPbBr₃ nanocrystals

Colloidal crystals of CsPbBr₃ were synthesized by the method of work [8], purified and dispersed in toluene. The surface of nanocrystals during the synthesis was covered with molecules of oleic acid and oleylamine. Although in solution these ligands provide such valuable properties as sedimentation resistance and passivation of surface states, the presence of these ligands is critical for precipitation. The general problem with the deposition of such colloidal NCs from solutions by dip-coating is that the following dipping results in partial or full dissolution of the previous layer due to colloidal nature of nanocrystals. To remove ligands, we used methyl acetate as antisolvent that removes excess of ligands [9] and results in deposition of material. To passivate the surface states due to the lead vacancies remaining when some of the ligands are removed, a small amount (<1 wt. %) of lead nitrate was added to the methyl acetate solution.

Thus, the procedure for depositing nanocrystals consists of successively immersing the substrate in a solution of nanocrystals and a solution of lead nitrate in methyl acetate. The number of repetitions determines the thickness of the resulting layer. To assess the layer deposition procedure, a cover slip was used as a test sample. As can be seen from Figure 2, a three-fold repetition of the procedure leads to a proportional increase in the optical density, which is commensurate with the thickness of the film. The spectrum feature near 530 nm corresponds to the first exciton peak of nanocrystals.

When the films were placed under an ultraviolet lamp, it was found that the deposition procedure leads to a significant decrease in the luminescence intensity in comparison with nanocrystals in solution or films obtained without ligands etching. However, the material still represented an interest in light absorbing.

3. Results and discussion

One of the well-known problems for hybrid perovskites is rapid structural degradation of the perovskite/zinc oxide contact. While in the case of titanium oxide such problem is not observed, zinc oxide deprotonates the organic component of hybrid perovskite such as $\text{CH}_3\text{NH}_3\text{PbI}_3$ [11]. Obviously, in the case of the CsPbX_3 system, deprotonation is excluded. Therefore, to test the proposed procedure, we deposited a perovskite CsPbBr_3 layer of nanocrystals on ZnO nanorods obtained by the hydrothermal method [12] on the surface of a compact ZnO layer deposited by ultrasonic spray pyrolysis. Contacts to the structure represent a transparent electrode of tin oxide doped with fluorine (FTO) and pinch gold contacts, used for the measurements. The corresponding structure is shown in Figure 3.

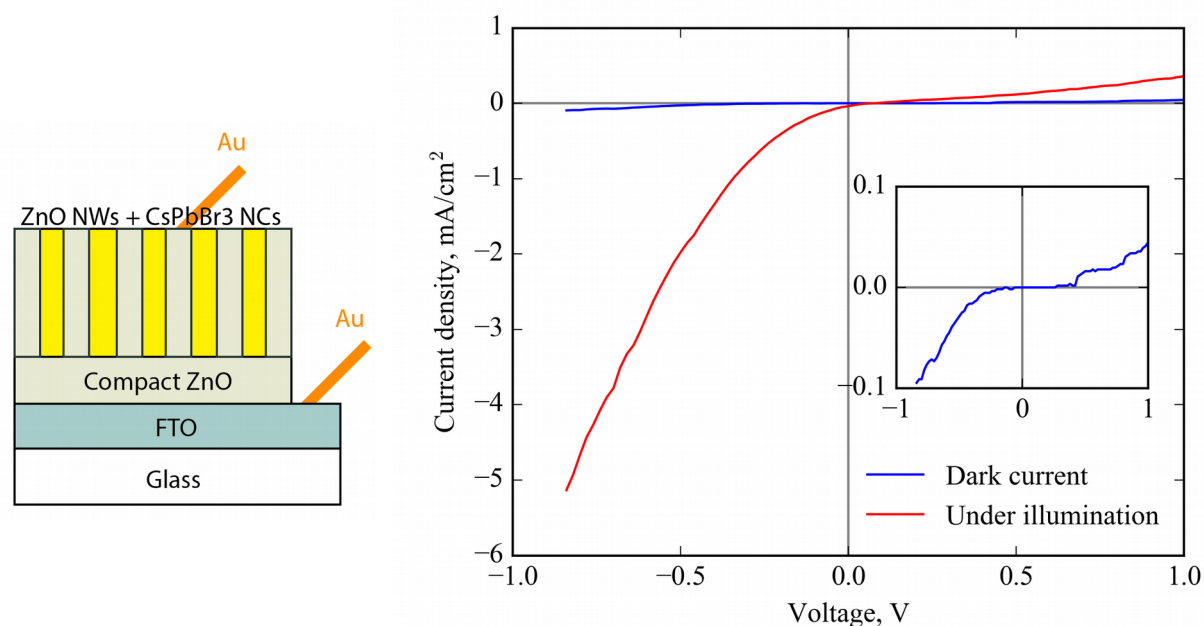


Figure 3. FTO/ZnO/CsPbBr₃/Au structure and corresponding current–voltage characteristic

The nanowires obtained by this method are fixed perpendicular to the substrate and have an extended structure ($\sim \mu\text{m}$). In fact, when measuring such structures, we investigated the effect of perovskite nanocrystals deposited on the surface of charge carriers inside zinc oxide rods. The corresponding current-voltage characteristics in the dark and under illumination are shown in Figure 3. It can be seen that when illuminated, the current through the structure increases approximately 50 times. Since the remaining materials of the structure are wide-gap and do not exhibit a strong photoresponse, it can be concluded that such a significant current change is due to charge injection from the perovskite layer to ZnO NWs. The effect is reversible, and no degradation of the sample was observed during the measurements.

4. Conclusion

Thus, a method for creating perovskite layers from colloidal solutions has been proposed, and the possibility of its application in the composition of a photosensitive structure based on a perovskite/zinc oxide contact is demonstrated. The same device is applicable for the creation of electron and hole transport layers [12]. Thus, this technique can be used for the synthesis of different layers of the final structure.

Acknowledgments

The authors thank Andrei Ryabko for providing a sample of zinc oxide for creating the described structure.

References

- [1] Protesescu L et al. 2015 *Nano Lett.* **15** 3692
- [2] Park, N G 2013 *J. Phys. Chem. Lett.* **4** 2423
- [3] Kulbak M, Cahen D, Hodes G 2015 *J. Phys. Chem. Lett.* **6** 2452
- [4] Nedelcu G, et al. 2015 *Nano Lett.* **15** 5635
- [5] Akkerman Q et al. 2015 *J. Am. Chem. Soc.* **137** 10276
- [6] Aleksandrova O, Matyushkin L, Hondryukov D 2015 Patent Rus, no. 2015114951
- [7] Andronov A O, Matyushkin L B, Hondryukov D V 2016 *Young Researchers in Electrical and Electronic Engineering Conference (EIConRusNW), 2016 IEEE NW Russia* pp 19–22
- [8] Protesescu, L et al. 2015 *Nano Letters* **15**(6) 3692
- [9] Swarnkar A et al. 2016 *Science* **354**(6308) 92
- [10] Yang J et al. 2015 *Chem. Mat.* **27**(12) 4229
- [11] Lashkova N A et al. 2016 *Semiconductors* **50**(9) 1254
- [12] Matyushkin L, Reshetnikova A et al. 2017 *Semiconductors* **5** 622